



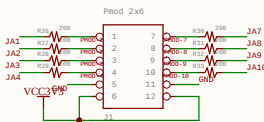
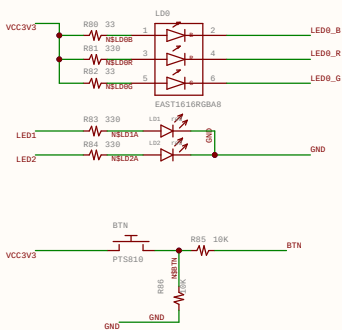
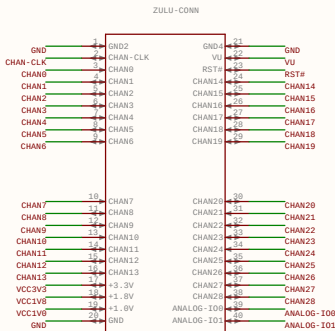
X2: 40 pins on a 2.54 mm grid, rows 22.86 mm (9.000 in) apart, numbered 1-40 straight through from the right and skipping both landings. The two rows are mirror images.

TOP ROW, x 59.69 down to 1.27: GND (pin 1), CHAN-CLK (2), CHAN0-CHAN6 (3-9), CHAN7-CHAN12 (10-16), CHAN13-CHAN14 (17-18), CHAN15-CHAN16 (19-20).

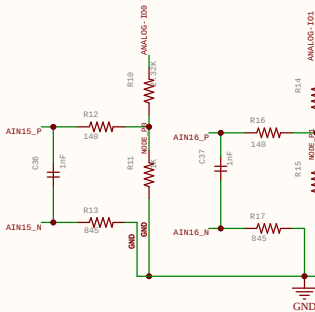
BOTTOM ROW, x 59.69 down to 1.27: GND, VIO, RSTa (pins 21-23), CHAN1A, CHAN1B (24-26), CHAN17...CHAN19 (27-29), then CHAN20, CHAN28 (30-38), ANALOG-IO0, ANALOG-IO1 (39, 40).

FOUR positions in each row - x 36.83, 34.29, 31.75, 29.21, between pins 19 and 38 above and 20 and 39 below are RESERVED for future use. These are spare pins.

micro-SB lands on the top row (7.80 mm of copper), top-side LPO header x4 (JST BSH pin-SM4-T2, 7.95 mm body) lands on the bottom row directly opposite it.



Digilent 12-pin Pmod (J1) Pins 1-4 and 7-10 use signals through 200R series resistors; pins 5/11 are GND and 6/12 are VCC3V3 -- so a 2x6 header carries EIGHT signals, not twelve, which is why the names skip JA5/JA6. No decoupling here: the Pmod spec puts bypassing on the module, and VCC3V3 already carries ~78uF. The Cmod A7 has none at its Pmod either. JA1-JA4 (the top row, the half a Pmod uses for SPI) are all in bank 16: A14 A15 B15 A16. JA7 is C15, also bank 16. JA8/JA9/JA10 are M19/M2/M7 -- banks 14/35/34 -- because nothing else was free. All five banks are VCC0=3.3V, so this costs timing coherence, not levels.



U3 = AS4C32M16SB-7TCN: 512Mbit (32M x 16) = 64MB.
Pin-for-pin and dimensionally identical to the 256MB
AS4C16M16SA-6TIN it replaces -- all 54 pins and every
TSP: 1 dimension checked against both datasheets, and
pin 40 is NC on both.
One DIFFERENCE, and IT IS NOT A PIN: the column address
is A0-A9 (1024 columns), not A0-A8 (512). The memory
controller must be set to 10 column bits, or it reaches
only half the array. Row A0-A12, banks BA0/BA1 and
refresh (8192 cycles / 64ms) are unchanged.
-7TCN since 2026-09-08 (-6TIN dry): 143 MHz max at CL13,
tRCD/trp 21, tRC 63 ns, TA 0-70 C, ID01 max 110 mA.

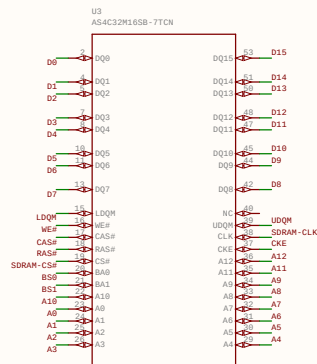


Figure 1 shows the schematic diagram of the proposed 8-bit DAC. The circuit is composed of eight identical stages, each corresponding to a bit of the 8-bit input. The inputs are labeled VSSQ[3] through VSSQ[9], and the outputs are labeled VDDQ[3] through VDDQ[9]. Each stage contains a 1-bit DAC core, which is a combination of a 1-bit DAC and a 1-bit DAC. The 1-bit DAC core is implemented using a 1-bit DAC (VCC3V3, 1.2uF, 1.2uF) and a 1-bit DAC (VCC3V3, 1.2uF, 1.2uF). The outputs of the 1-bit DACs are summed to produce the final 8-bit DAC output.

Memory			
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